

Two Channels for $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ Loop Conversion in RadCluster

Theoretical basis, kinetic derivation, and interpretation

Technical working note

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Abstract

RadCluster treats the observed $\frac{1}{2}a\langle 111 \rangle$ and $a\langle 100 \rangle$ loop populations in ferritic/martensitic steels as two coupled cluster-dynamics populations. Conversion is represented by two physically distinct channels: (A) a size-preserving, thermally activated unary conversion whose driving force is obtained from the anisotropic-elastic free energies of Dudarev, Bullough and Derlet; and (B) the collision-mediated Marian–Wirth–Perlado mechanism, comprising binary junction nucleation and subsequent growth of a sessile $\langle 100 \rangle$ loop by capture of mobile $\frac{1}{2}\langle 111 \rangle$ loops. The channels are complementary rather than redundant. The Dudarev physics determines which character is thermodynamically preferred and produces the strong temperature selection; the Marian physics supplies a geometrically plausible pathway for creating and growing large $\langle 100 \rangle$ loops. This note derives the equations used in RadCluster, identifies the modeling closures that do not follow uniquely from the source papers, and states what an ablation study must demonstrate before claiming that both channels are quantitatively necessary.

1 State variables and the physical question

Let c_n^{111} and c_n^{100} be number densities of SIA clusters containing n interstitials and having, respectively, $\frac{1}{2}a\langle 111 \rangle$ and $a\langle 100 \rangle$ Burgers vectors. Cascade production primarily feeds the 111 ladder. The 111 loops are glissile and migrate approximately one-dimensionally, with direction changes producing effective 1D/3D transport; the 100 loops are treated as sessile. The conversion problem is therefore not merely a choice of loop energy. A cluster-dynamics model must specify

$$\text{thermodynamic preference} + \text{transition pathway} + \text{encounter kinetics}. \quad (1)$$

The Dudarev and Marian papers constrain different pieces of this statement. Dudarev *et al.* explain the temperature-dependent free-energy ordering of loop characters; they do not prescribe a unique population-level conversion rate. Marian *et al.* provide an atomistic collision pathway and activation landscape; they do not by themselves supply the full temperature-dependent loop free energy used to reproduce the ferritic-steel crossover. Treating the two as additive CD channels is therefore defensible, provided their overlap is calibrated without double counting.

2 Channel A: Dudarev thermodynamics made into a unary kinetic law

2.1 Loop energy and the origin of the temperature crossover

For a prismatic loop of character $X \in \{111, 100\}$, the anisotropic-elastic line-energy representation may be written

$$E_X(n, T) = P_X(n) \left[\hat{F}_X(T) \ln \left(\frac{4R_X(n)}{e\delta} \right) + F_{\delta, X}(T) + F_{c, X} \right], \quad (2)$$

where P_X is perimeter, R_X an equivalent radius, δ a core cutoff, $\hat{F}_X$ the anisotropic prelogarithmic factor, and $F_\delta + F_c$ the core-traction and nonlinear-core contributions. The platelet area follows from conservation of inserted atomic volume,

$$A_X(n)b_X = n\Omega, \text{quad} R_X(n) = \sqrt{\frac{n\Omega}{\pi b_X}}, \quad (3)$$

so $P_X \propto R_X \propto \sqrt{n}$ for a geometrically similar family of loops.

The crucial result of Dudarev *et al.* is that the pure-edge 001[100] prelogarithmic factor contains the soft shear modulus $c' = (c_{11} - c_{12})/2$:

$$\widehat{F}_{001}([100]) = \frac{a^2}{4\pi}(c_{11} + c_{12}) \left[\frac{c_{44}(c_{11} - c_{12})}{c_{11}(c_{11} + c_{12} + 2c_{44})} \right]^{1/2}. \quad (4)$$

Thus $\widehat{F}_{100} \rightarrow 0$ as $c' \rightarrow 0$ on approaching the bcc–fcc transition, while the corresponding 111 factors remain finite. If $c'(T) \propto (1 - T/T_c)^{1/2}$, then Eq. (4) gives the characteristic quarter-power scaling

$$\widehat{F}_{100}(T) \propto (1 - T/T_c)^{1/4}. \quad (5)$$

The thermodynamic force for conversion at fixed SIA content is

$$\Delta F(n, T) = E_{111}(n, T) - E_{100}(n, T). \quad (6)$$

Positive ΔF means the 100 loop is the lower-free-energy state. Equations (2)–(6), rather than a phenomenological label “Dudarev conversion,” are the theoretical basis of Channel A.

2.2 From free-energy difference to a rate

Transition-state theory gives a forward rate

$$k_{111 \rightarrow 100} = \nu_0 \exp \left[-\frac{F^\ddagger - F_{111}}{k_B T} \right]. \quad (7)$$

For a reversible two-state model, microscopic consistency requires

$$\frac{k_{111 \rightarrow 100}}{k_{100 \rightarrow 111}} = \exp \left(\frac{\Delta F}{k_B T} \right). \quad (8)$$

Neither Eq. (7) nor the Dudarev free energies determine the saddle free energy $F^\ddagger$. RadCluster therefore introduces a kinetic closure. Its one-way unary rate has the form

$$\Gamma_{\text{uni}}(n, T) = \nu_0 \exp \left[-\frac{E_a^{\text{uni}}(n)}{k_B T} \right] \left[1 - \exp \left(-\frac{\Delta F(n, T)}{k_B T} \right) \right]_+, \quad (9)$$

where $[x]_+ = \max(0, x)$. The second factor is a thermodynamic gate. It vanishes at coexistence, is zero when 100 is unfavorable, and approaches unity for a strongly downhill transformation. It may be understood as the normalized net forward-minus-reverse tendency when rates obey Eq. (8); it is not a unique consequence of Dudarev’s theory.

The implemented barrier is perimeter dependent,

$$E_a^{\text{uni}}(n) = E_a^0 + \gamma_a \frac{P(n)}{b_{111}}, \quad (10)$$

motivated by a conversion that propagates through successive dislocation segments. Since $P \sim \sqrt{n}$, Eq. (10) suppresses direct rotation of large loops. At the same time, $\Delta F \sim P \sim \sqrt{n}$ strengthens the downhill gate. Their competition creates a finite size window for unary conversion. This size dependence is a RadCluster modeling hypothesis that must be calibrated; it should not be presented as a direct derivation from the Dudarev paper.

In the CD equations, this channel is a size-preserving population transfer:

$$\dot{c}_n^{111} \Big|_{\text{uni}} = -\Gamma_{\text{uni}}(n, T) c_n^{111}, \quad (11)$$

$$\dot{c}_n^{100} \Big|_{\text{uni}} = +\Gamma_{\text{uni}}(n, T) c_n^{111}. \quad (12)$$

Consequently $\sum_n n(c_n^{111} + c_n^{100})$ is conserved exactly by Channel A.

3 Channel B: Marian collision, junction formation, and templated growth

3.1 Encounter kernel

For two dilute mobile objects, the classical Smoluchowski result is the steady diffusive flux to an absorbing relative-coordinate surface:

$$K_{nn'}^{3D} = 4\pi(R_n + R_{n'})(D_n + D_{n'}). \quad (13)$$

When concentrations are stored per atomic volume and radii are nondimensionalized, this produces the code's conventional factor proportional to

$$K_{nn'}^{ii} = \frac{8\pi}{\Omega^{2/3}}(\xi_n + \xi_{n'})(D_n^{\text{eff}} + D_{n'}^{\text{eff}}), \quad \xi_n = \left(\frac{3n}{8\pi}\right)^{1/3}. \quad (14)$$

For $\frac{1}{2}\langle 111 \rangle$ loops, D^{eff} is not simply the 1D glide coefficient. It is an isotropic-equivalent coefficient representing a glider that occasionally changes glide direction. Thus Eq. (14) is a mean-field encounter closure for 1D/3D migration, not an exact first-passage solution for two crystallographic lines. This distinction matters because the Marian channel scales directly with encounter frequency.

3.2 Junction nucleation

Marian *et al.* identified an atomistic reaction between suitably oriented, comparably sized $\frac{1}{2}\langle 111 \rangle$ loops, schematically

$$\frac{1}{2}[111] + \frac{1}{2}[\bar{1}\bar{1}\bar{1}] \longrightarrow [100]. \quad (15)$$

The SIA content of the product is $n + n'$. RadCluster partitions every 111–111 encounter into an ordinary 111 coalescence branch and a 100-forming branch:

$$K_{nn'}^{\text{junc}} = \phi_{nn'} K_{nn'}^{ii}, \quad K_{nn'}^{111 \text{ coal}} = (1 - \phi_{nn'}) K_{nn'}^{ii}. \quad (16)$$

The branching fraction encodes the atomistic requirements that the reactants be large enough and similar in size:

$$\phi_{nn'} = \phi_{\text{max}} \exp\left[-\frac{\ln^2(n/n')}{2\sigma_s^2}\right] \Theta(\min(n, n') - n_{j,\text{min}}) P_{\text{suc}}^j(T). \quad (17)$$

The log-Gaussian is symmetric under $n \leftrightarrow n'$, peaks for equal sizes, and depends on a size ratio rather than an arbitrary absolute difference. It is a useful coarse-grained surrogate for orientation, impact parameter and size compatibility; it is not an atomistically calculated probability.

The Marian energy landscape contains a metastable $\frac{1}{2}\langle 110 \rangle$ intermediate. Once that intermediate exists, suppose completion to 100 and reversal to 111 are independent activated Poisson processes,

$$k_2 = \nu_2 e^{-\Delta H_2/k_B T}, \quad k_{\text{rev}} = \nu_r e^{-\Delta H_{\text{rev}}/k_B T}. \quad (18)$$

The probability that completion occurs first is the standard competing-hazards result

$$P_{\text{suc}}^j(T) = \frac{k_2}{k_2 + k_{\text{rev}}} = \frac{A_j e^{-\Delta H_2^j/k_B T}}{A_j e^{-\Delta H_2^j/k_B T} + e^{-\Delta H_{\text{rev}}^j/k_B T}}, \quad (19)$$

where $A_j = \nu_2/\nu_r$. This derivation explains why the success factor multiplies the encounter branching fraction rather than the diffusivity: diffusion creates the junction opportunity; the activated competition determines whether that opportunity produces a persistent 100 loop.

The corresponding gain-loss terms are

$$\dot{c}_n^{111}|_{\text{junc}} = -c_n^{111} \sum_{n'} K_{nn'}^{\text{junc}} c_{n'}^{111}, \quad (20)$$

$$\dot{c}_m^{100}|_{\text{junc}} = \frac{1}{2} \sum_{n+n'=m} K_{nn'}^{\text{junc}} c_n^{111} c_{n'}^{111}. \quad (21)$$

The factor 1/2 prevents double counting unordered pairs. Multiplying by size and summing shows that the 111 loss is exactly the 100 gain in SIA inventory.

3.3 Growth by capture of mobile 111 loops

After nucleation, a sessile 100 loop can grow by capturing a mobile 111 loop:

$$[100]_m + \frac{1}{2}\langle 111 \rangle_n \longrightarrow [100]_{m+n}. \quad (22)$$

Because $D_m^{100} \simeq 0$, the mean-field cross-character rate is

$$K_{mn}^{\text{abs}} = P_{\text{suc}}^{\text{abs}}(T) B_{\text{abs}} \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) D_n^{\text{eff}}. \quad (23)$$

Here B_{abs} is a capture enhancement or calibration factor. Current RadClusterdesign notes allow the absorption success gate to have its own barrier and prefactor, separate from junction nucleation. That separation is physically sensible: a pre-existing 100 loop templates the final character, whereas two 111 loops must first create a viable 100 nucleus. The equations are

$$\dot{c}_n^{111}|_{\text{abs}} = -c_n^{111} \sum_m K_{mn}^{\text{abs}} c_m^{100}, \quad (24)$$

$$\dot{c}_m^{100}|_{\text{abs}} = \sum_n K_{m-n,n}^{\text{abs}} c_{m-n}^{100} c_n^{111} - c_m^{100} \sum_n K_{mn}^{\text{abs}} c_n^{111}. \quad (25)$$

With appropriate boundary handling, Eqs. (24)–(25) also conserve total SIA content.

4 Why both channels can be needed—and what is actually demonstrated

The mechanisms occupy different regions of kinetic phase space:

	Dudarev-based unary channel	Marian junction/absorption channel
Primary basis	Temperature-dependent anisotropic loop free energy	Atomistic collision, intermediate junction, and reorientation pathway
Reaction order	First order in c_n^{111}	Second order for nucleation; bilinear for subsequent absorption
Main selectivity	Strong thermodynamic temperature gate; barrier suppresses large-loop rotation	Requires encounters, compatible sizes/orientations, and successful completion; preferentially builds larger loops
Natural role	Establishes the low- T /111 to high- T /100 crossover	Seeds 100 loops and preserves/grows them toward observable sizes with dose
Key uncertainty	Saddle barrier and its size scaling	Encounter dimensionality, branching probability, barriers, and capture enhancement

The model behavior recorded in the repository is consistent with this division. An early ungated Marian implementation converted loops too efficiently at all temperatures and masked the Dudarev trend. Introducing the competing-hazard success probability restored an activated kinetic bottleneck. Conversely, a unary barrier that is too low also causes excessive conversion; when the barrier grows with perimeter, the unary route becomes biased toward small loops and can yield a peak followed by decay in the 100 fraction as the population coarsens beyond its conversion window. Marian absorption then provides a route for dose-persistent growth of already nucleated 100 loops.

This supports the qualitative statement

Dudarev selects thermodynamic character with temperature; Marian realizes nucleation and large-loop growth kinetically.

It does *not* yet prove that both are quantitatively necessary for one fixed parameter set. In particular, the repository calibration notes indicate that a sufficiently large second-step barrier can make P_{suc} extremely small in the relevant temperature range, effectively switching off the Marian channel. A claim of necessity requires converged, dose-matched calculations with identical non-conversion parameters:

1. neither conversion channel;
2. unary Dudarev-based conversion only;
3. Marian junction plus absorption only;
4. both channels together.

The decisive observables are not only the total 100 fraction, but also N_{111} , N_{100} , their mean diameters and full size distributions versus both temperature and dose. Channel A should chiefly recover temperature selection; Channel B should chiefly recover nucleation/growth kinetics and persistence. If only the combined case reproduces both classes of observables with one parameter set, the assertion that both mechanisms are necessary becomes well supported.

5 Implementation correspondence and cautions

The present repository declares three separate reaction-graph edges:

- `loop_111to100_unary`, with kernel `Gamma_uni`;
- `SIA111_junction`, with kernel `K_111_junction`;
- `SIA100_absorb`, with kernel `K_100_absorb`.

This separation is scientifically valuable because it permits a clean ablation and retains exact SIA stoichiometry.

Three cautions should accompany publication of the model. First, the unary kinetic equation is a `RadClusterclosure` built on Dudarev thermodynamics, not an equation derived in the 2008 paper. Second, the log-Gaussian junction yield and absorption enhancement are mesoscale closures for unresolved geometry. Third, using an isotropic-equivalent diffusivity for 1D/3D loop motion can under- or over-estimate loop-loop and loop-sink encounters; its sensitivity must be separated from conversion-barrier sensitivity. These qualifications strengthen the analysis by locating precisely what is theoretical input, what is atomistic evidence, and what must be calibrated.

References

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